

CMPDI DETAILED PROJECT REPORT (DPR) - Jhanjra Underground Continuous Miner

Subsidiary: ECL | Capacity: 5.2 MT | Geological Basin: Raniganj Coalfield, West Bengal

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This Detailed Project Report (DPR) formulates the mine planning, equipment sizing, and geological stripping layout for Jhanjra Underground Continuous Miner under ECL.

Geological Reserve & Setting: Situated in the Raniganj Coalfield, West Bengal, the mine exploits thick coal seams belonging to the Barakar Formation. Total geological reserves are estimated at 480 Million Tonnes.

Rated Production Capacity: 5.2 MT under mechanized Underground operations. Stripping ratio is projected at Continuous Miner & Longwall, utilizing 42 m3 electric rope shovels, 240T rear dump trucks, and high-capacity bucket wheel systems.

Environmental Mitigation & FMC: Coal evacuation is planned via a First Mile Connectivity (FMC) rapid loading silo with in-pit crushing and a 4.2 km overland conveyor to eliminate surface haul truck emissions.

Technical Parameter	Project Specification	DGMS / Ministry Compliance
Subsidiary / Mine Name	ECL - Jhanjra_Underground_	Approved by CIL Board
Rated Annual Capacity	5.2 MT	Environmental Clearance Granted
Mining Method	Underground	High-wall Stability Factor > 1.3
Stripping Ratio	Continuous Miner & Longwall	Continuous Reclamation Mandated
Evacuation Mode	Rapid Loading Silo / FMC	Zero Surface Dust Spillage